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**Agile Development Process and Devops Lab – ISWE406P**

**Final Assessment Test**

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**Slot:** L3 + L4

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**Register Number:** 23MIS0536

## SET – 1

### Online Food Ordering System

### Problem Statement:

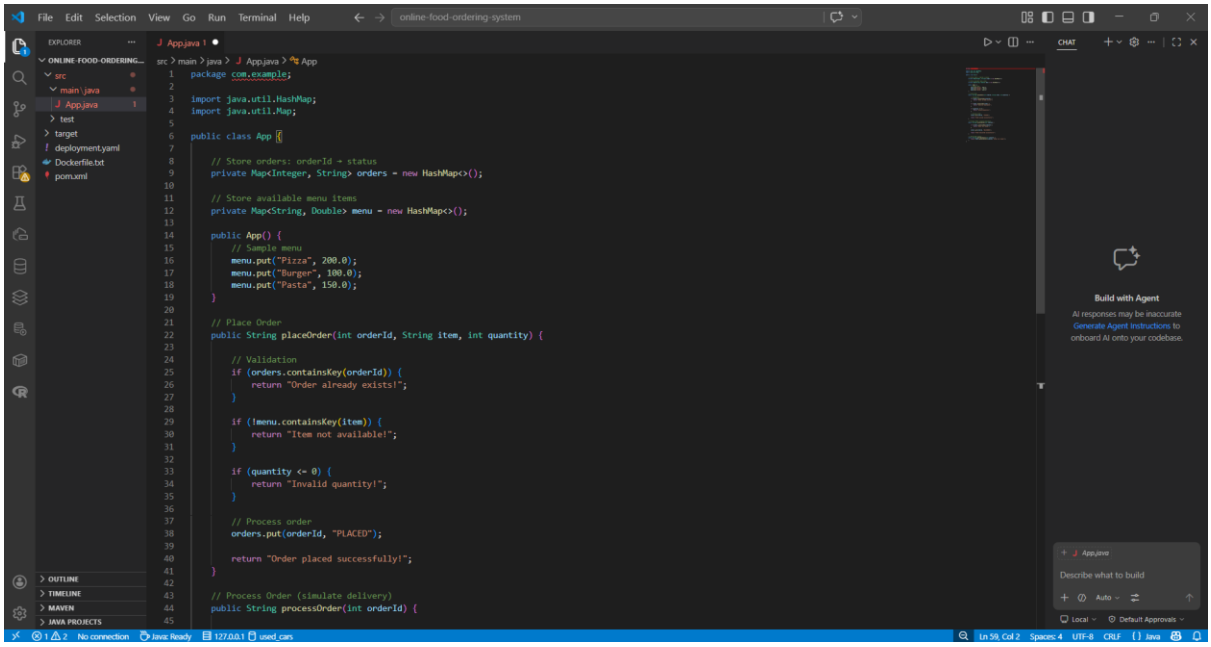
Ensure food orders are placed and processed correctly without errors.

### Tasks:

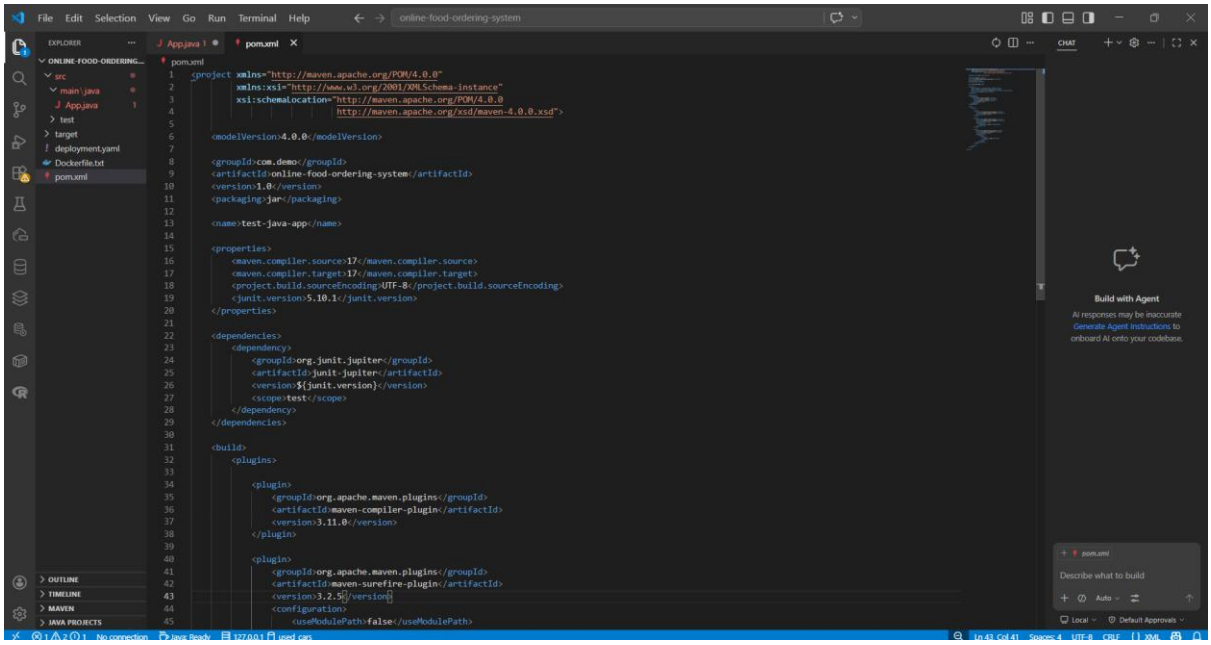
1. Develop order placement and processing logic
2. Write JUnit tests for order success/failure scenarios
3. Implement CI/CD pipeline with automated testing
    - Set up version control using GitHub
    - Configure CI pipeline(e.g., Jenkins/GitHub Actions) to trigger on code commits
    - Automate build process using Maven
    - Integrate automated unit tests in pipeline
    - Configure CD pipeline for automatic deployment to staging/production
4. Deploy services on Docker and Kubernetes
    - Containerize application using Docker(write Dockerfile)
    - Build and push Docker image to container registry (Docker Hub)
    - Create Kubernetes deployment and service YAML files

**Step 1:** Develop order placement and processing logic

### Create App.java



## Create pom.xml




```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
                             http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.demo</groupId>
    <artifactId>online-food-ordering-system</artifactId>
    <version>1.0</version>
    <packaging>jar</packaging>

    <name>test-java-app</name>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
        <junit.version>5.10.1</junit.version>
    </properties>

    <dependencies>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter</artifactId>
            <version>${junit.version}</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
        <plugins>

            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.11.0</version>
            </plugin>

            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-surefire-plugin</artifactId>
                <version>3.2.5</version>
                <configuration>
                    <useModulePath>false</useModulePath>
```

## Build Locally using Maven (Test)

## Inside project folder:

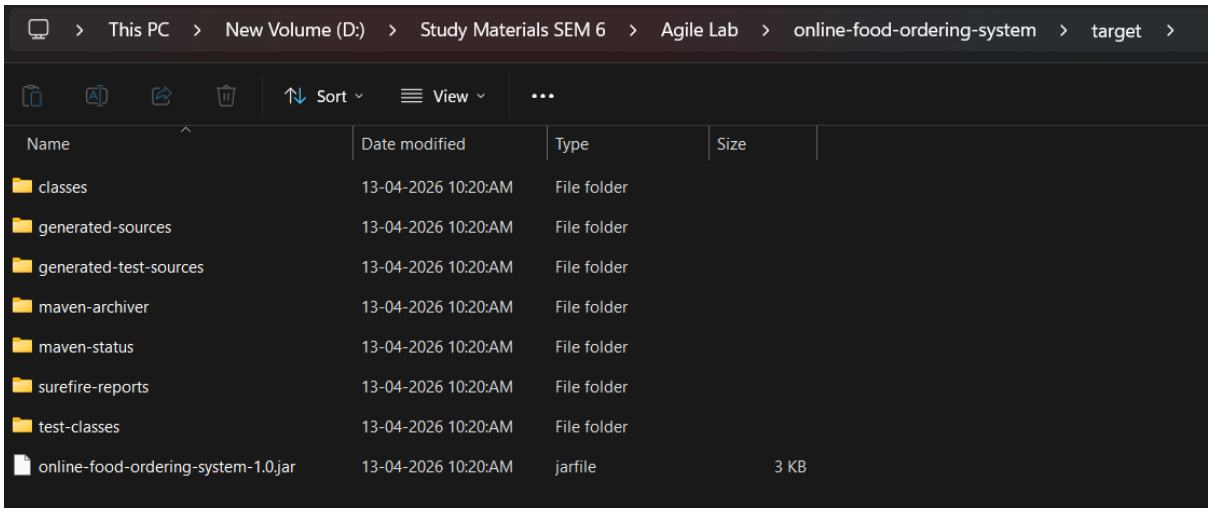
mvn clean test

```
PS D:\Study Materials SEM 6\Agile Lab\online-food-ordering-system> mvn clean test
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/platform/junit-platform-launcher/1.10.1/junit-platform-launcher-1.10.1.pom (3.0 kB at 39 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/platform/junit-platform-launcher/1.10.1/junit-platform-launcher-1.10.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/platform/junit-platform-launcher/1.10.1/junit-platform-launcher-1.10.1.jar (184 kB at 704 kB/s)
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.035 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.466 s
[INFO] Finished at: 2026-04-13T10:19:23+05:30
[INFO] ------------------------------------------------------------------------
PS D:\Study Materials SEM 6\Agile Lab\online-food-ordering-system>
```

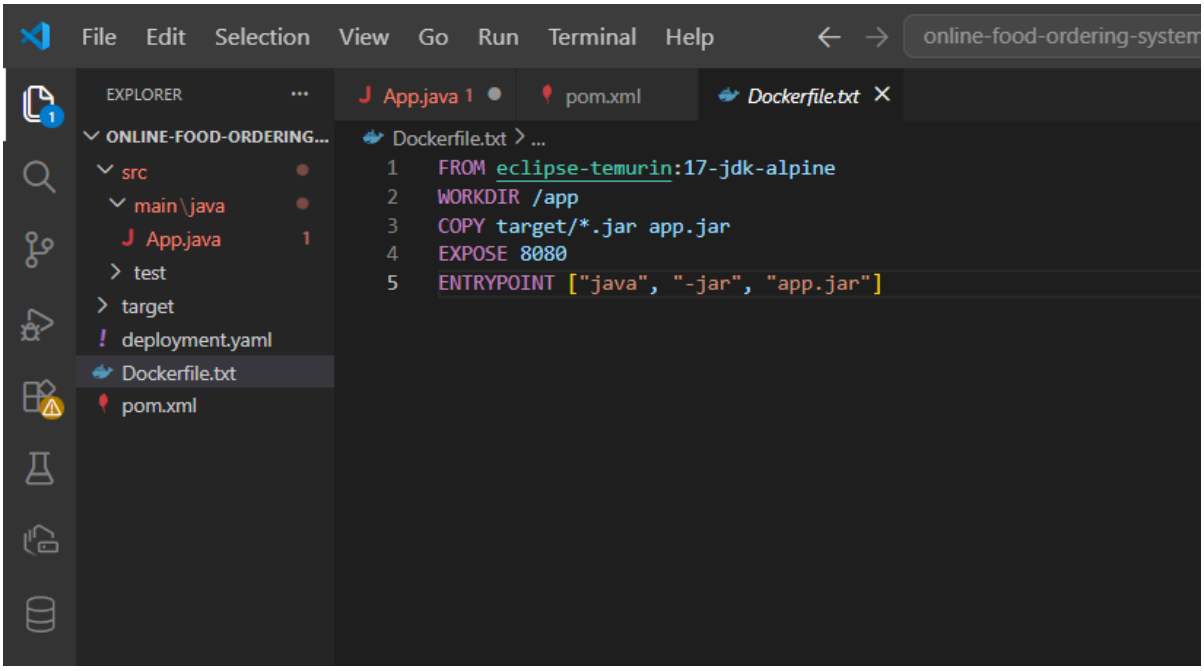
mvn clean package

```
PS D:\Study Materials SEM 6\Agile Lab\online-food-ordering-system> mvn clean package
[INFO] --- surefire:3.2.5:test (default-test) @ online-food-ordering-system ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.033 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ online-food-ordering-system ---
[INFO] Building jar: D:\Study Materials SEM 6\Agile Lab\online-food-ordering-system\target\online-food-ordering-system-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.246 s
[INFO] Finished at: 2026-04-13T10:20:41+05:30
[INFO] ------------------------------------------------------------------------
```

It generates: target/ online-food-ordering-system-1.0.jar

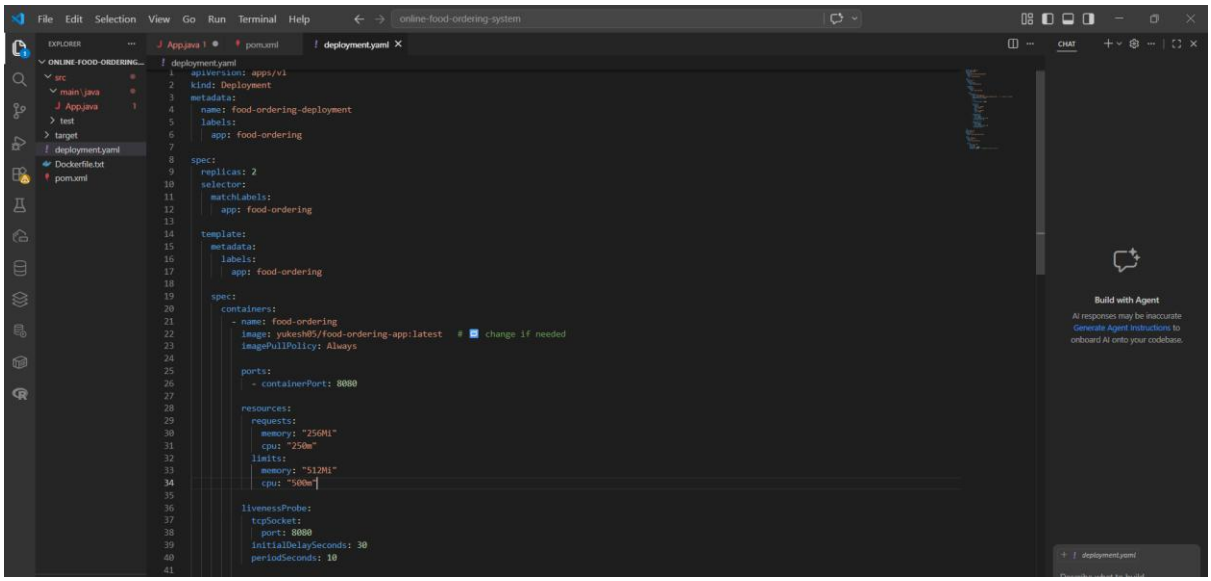



**Create Dockerfile**



**Create Kubernetes Deployment File**

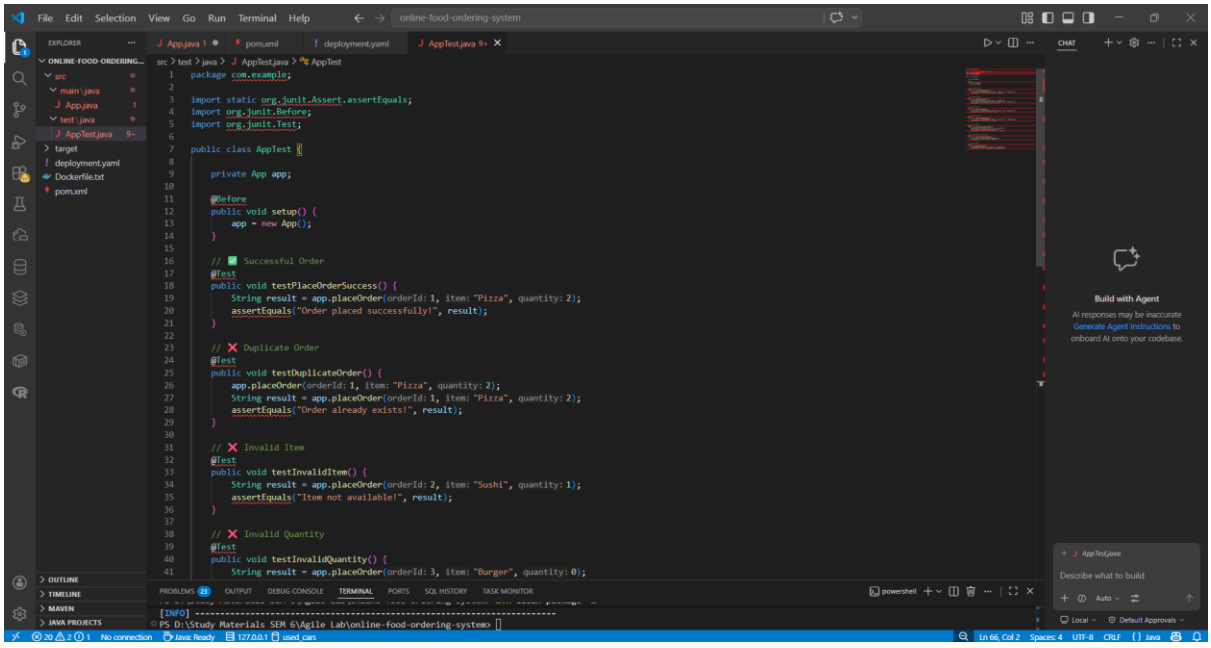
**deployment.yaml**



```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: food-ordering-deployment
  labels:
    app: food-ordering

spec:
  replicas: 2
  selector:
    matchLabels:
      app: food-ordering

  template:
    metadata:
      labels:
        app: food-ordering

    spec:
      containers:
        - name: food-ordering
          image: yukesh05/food-ordering-app:latest   # ✅ change if needed
          imagePullPolicy: Always

          ports:
            - containerPort: 8080

          resources:
            requests:
              memory: "256Mi"
              cpu: "250m"
            limits:
              memory: "512Mi"
              cpu: "500m"

          livenessProbe:
            tcpSocket:
              port: 8080
            initialDelaySeconds: 30
            periodSeconds: 10
```

**Step 2:** Write JUnit tests for order success/failure scenarios

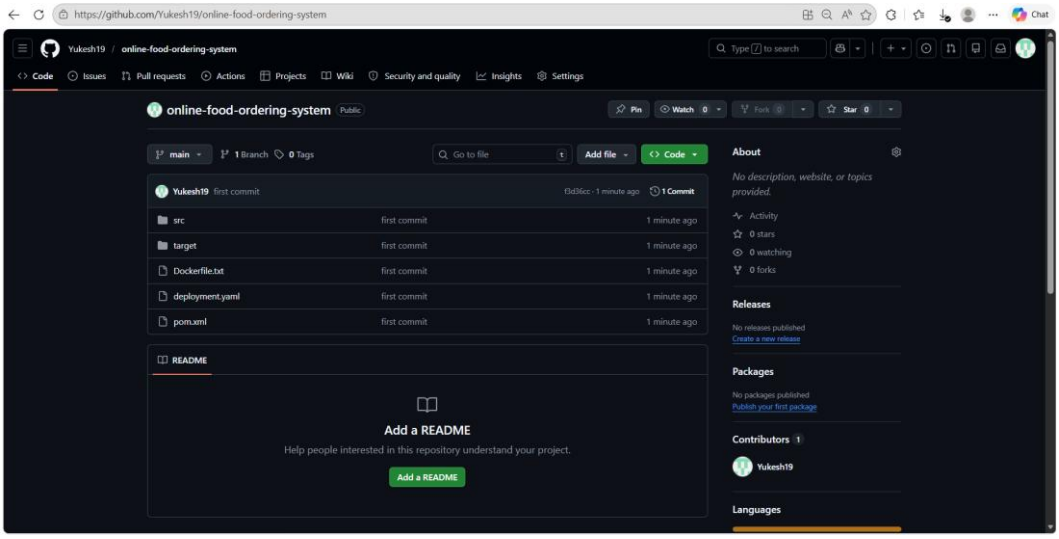
**Create AppTest.java**



```java
package com.example;

import static org.junit.Assert.assertEquals;
import org.junit.Before;
import org.junit.Test;

public class AppTest {

    private App app;

    @Before
    public void setup() {
        app = new App();
    }

    // ✅ Successful Order
    @Test
    public void testPlaceOrderSuccess() {
        String result = app.placeOrder(orderId: 1, item: "Pizza", quantity: 2);
        assertEquals("Order placed successfully!", result);
    }

    // ❌ Duplicate Order
    @Test
    public void testDuplicateOrder() {
        app.placeOrder(orderId: 1, item: "Pizza", quantity: 2);
        String result = app.placeOrder(orderId: 1, item: "Pizza", quantity: 2);
        assertEquals("Order already exists!", result);
    }

    // ❌ Invalid Item
    @Test
    public void testInvalidItem() {
        String result = app.placeOrder(orderId: 2, item: "Sushi", quantity: 1);
        assertEquals("Item not available!", result);
    }

    // ❌ Invalid Quantity
    @Test
    public void testInvalidQuantity() {
        String result = app.placeOrder(orderId: 3, item: "Burger", quantity: 0);
```

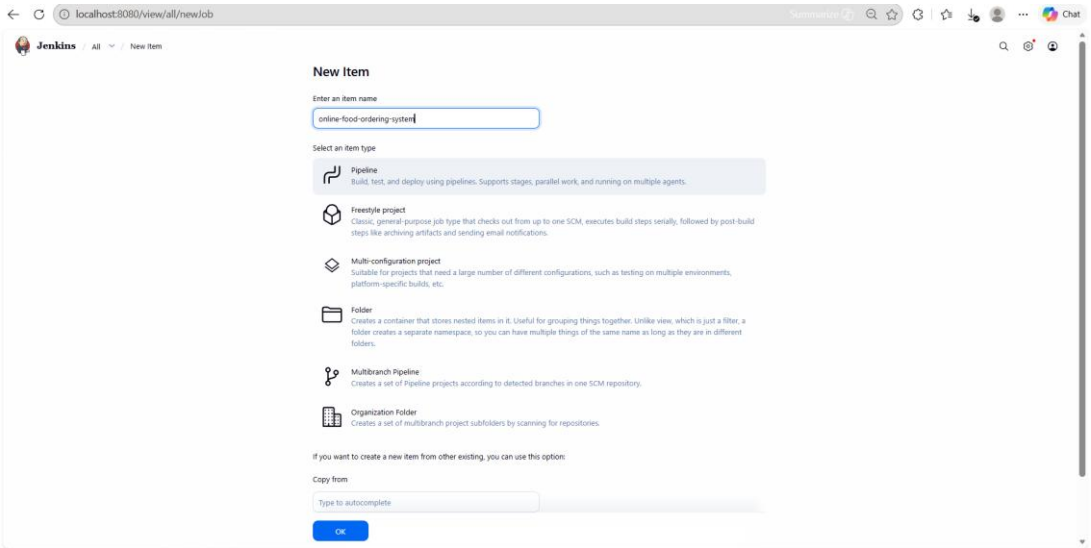
## Push code to GitHub

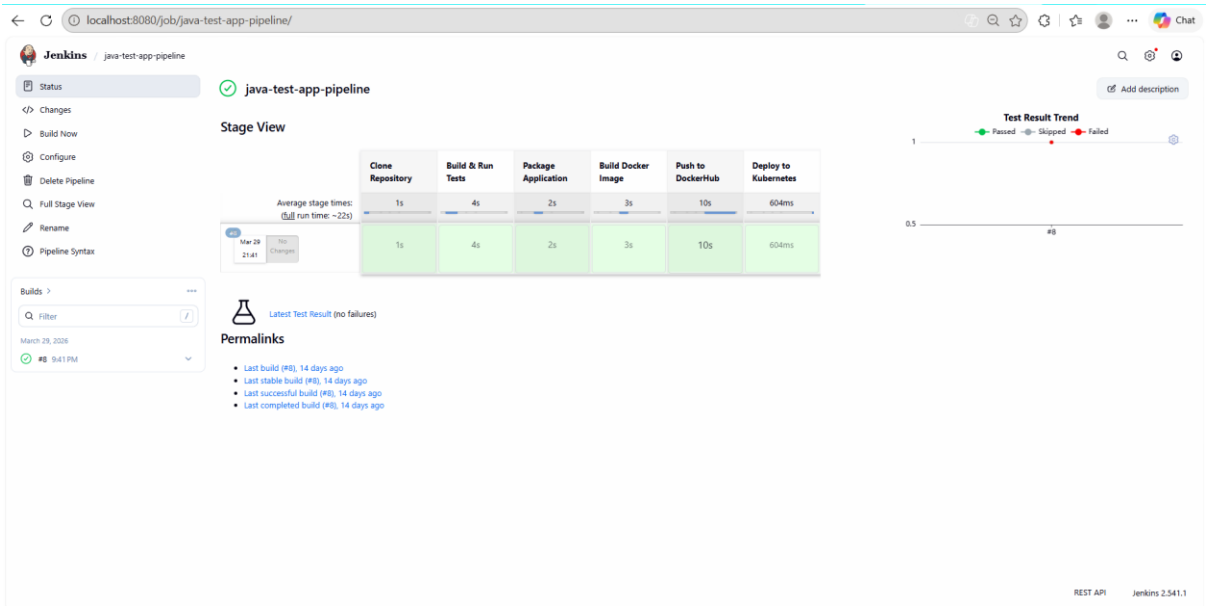
```
PS D:\Study Materials SEM 6\Agile Lab\online-food-ordering-system> git commit -m "first commit"
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/online-food-ordering-system-1.0.jar
 create mode 100644 target/surefire-reports/2026-04-13T10-20-40_299.dumpstream
 create mode 100644 target/surefire-reports/AppTest.txt
 create mode 100644 target/surefire-reports/TEST-AppTest.xml
 create mode 100644 target/test-classes/AppTest.class
PS D:\Study Materials SEM 6\Agile Lab\online-food-ordering-system> git branch -M main
PS D:\Study Materials SEM 6\Agile Lab\online-food-ordering-system> git remote add origin https://github.com/Yukesh19/online-food-ordering-system.git
PS D:\Study Materials SEM 6\Agile Lab\online-food-ordering-system> git push -u origin main
Enumerating objects: 34, done.
Counting objects: 100% (34/34), done.
Delta compression using up to 12 threads
Compressing objects: 100% (22/22), done.
Writing objects: 100% (34/34), 8.32 KiB | 851.00 KiB/s, done.
Total 34 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Yukesh19/online-food-ordering-system.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS D:\Study Materials SEM 6\Agile Lab\online-food-ordering-system>
```




**Step 3:** Implement CI/CD pipeline with automated testing

## Create Jenkins Pipeline Job

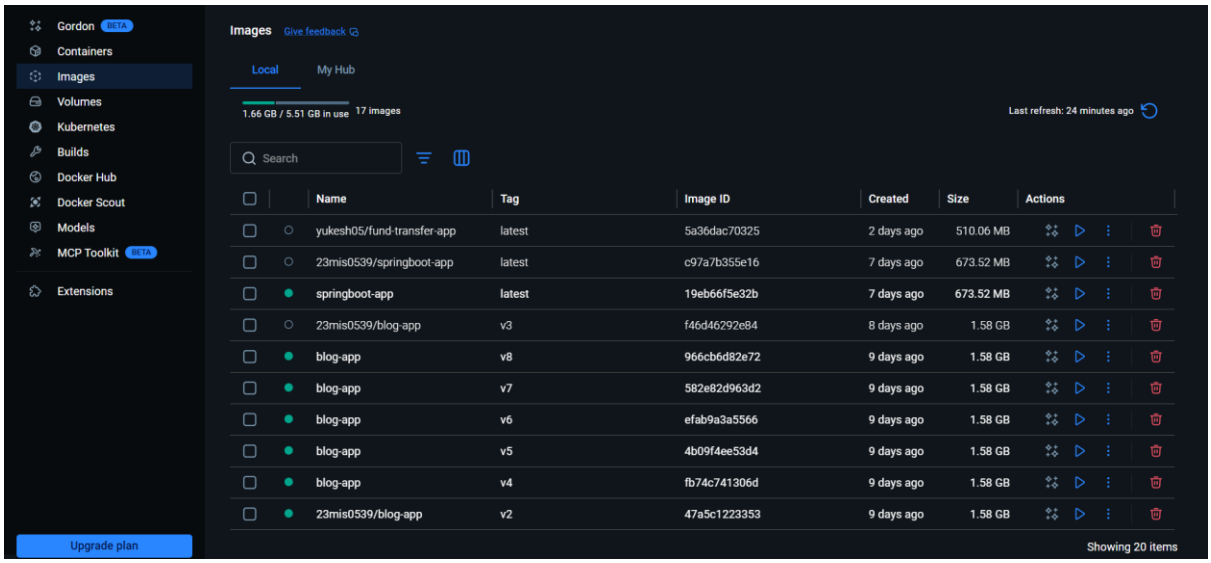


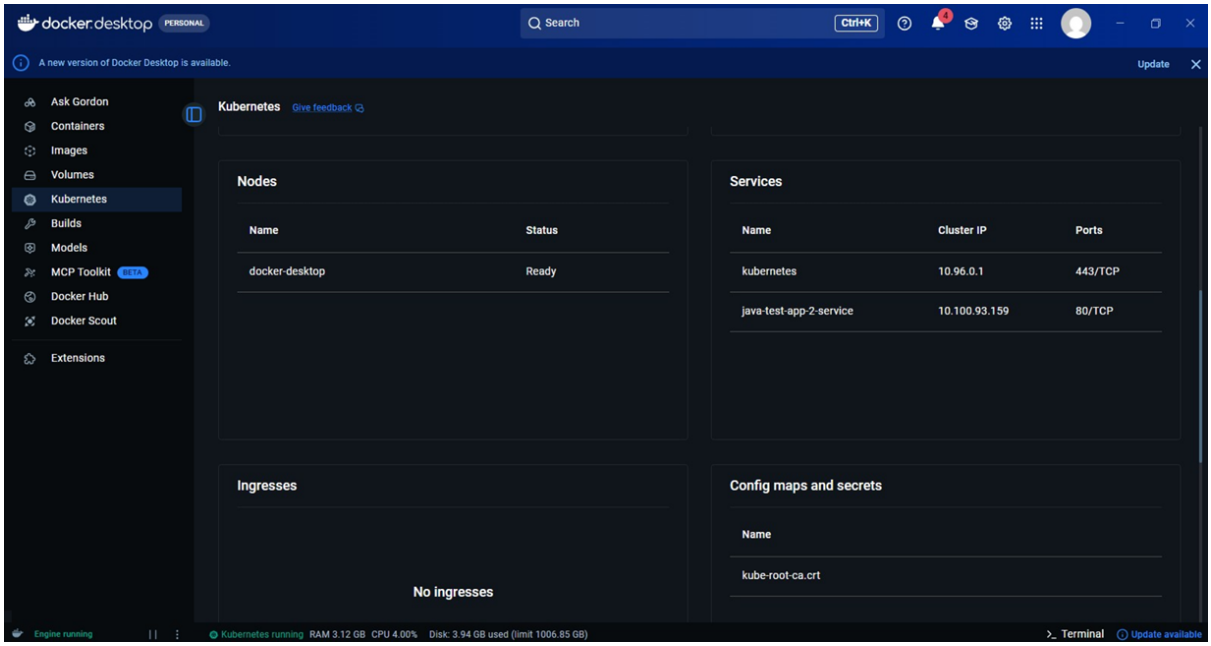



## Step 4: Deploy micro service on Docker and Kubernetes

## Verify Docker and Kubernetes Deployment

## Docker Image






```
C:\Users\ryuke>kubectl get svc
NAME                   TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
java-test-app-service  NodePort    10.96.191.227   <none>        80:30007/TCP   5m29s
kubernetes             ClusterIP   10.96.0.1       <none>        443/TCP        144m
```